MIL-F-19488(SHIPS)
27 July 1956

MILITARY SPECIFICATION

FITTINGS, HOSE, WATER, AND STEAM

(OTHER THAN FLEXIBLE, METALLIC)

1. SCOPE

1.1 Scope.- This specification covers hose fittings for attaching to various hoses.

1.2 Classification.- Hose fittings shall be furnished in the following types and classes, as specified (see 6.1):

Type I - Couplings.
  Class a - For cotton rubber-lined fire hose, wash deck and engineers rubber hose.
  Class b - For rubber steam hose.
  Class c - For water-suction hose.
Type II - Nozzles.
Type III - Strainers.
Type IV - Reducing couplings.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-P-775 - Packaging and Packing for Overseas Shipment and Domestic Shipment Involving Storage of Hose, Rubber or Fabric, (Including Tubing); and Fittings, Nozzles, and Strainers.
MIL-R-2765 - Rubber Material, Synthetic, Oil Resistant (Sheet, Strip, and Molded Shapes).

FSC 4210, 5120

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NAVY DEPARTMENT
General Specifications for Inspection for Material.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Material shall be of best quality and all fittings shall be free from porosity or other injurious defects.

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3.1.1 Chemical composition.- The chemical composition of the fittings shall be either A or B as specified in table I.

Table I - Chemical composition.

Composition	Copper	Tin	Lead	Iron (max.)	Nickel (max.)	Aluminum	Zinc
	Percent	Percent	Percent	Percent	Percent	Percent	Percent
A	82.0 - 84.0	2.5 - 3.5	2.5 - 3.5	0.35	.50	-----	Remainder
B	84.0 - 86.0	4.0 - 6.0	4.0 - 6.0	0.25	1.0	-----	Remainder

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3.1.2 Expansion ring for type I fittings.- Type I fittings shall have expansion rings cut from seamless drawn brass tubing. The chemical composition shall conform to table II.

Table II - Chemical composition of expansion rings.

Copper	Lead (max.)	Iron (max.)	Zinc
Percent	Percent	Percent	Percent
83.0 - 86.0	0.06	0.05	Remainder

3.1.3 Washer material.- The washer, part 5 of figure 1 shall be made of an oil resisting rubber in accordance with Specification MIL-R-2765.

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3.2 Design.-

3.2.1 Couplings.- All couplings shall conform to dimensions shown in tables III and IV. The length of the hose coupling threads shall be as shown on figure 2 and tables V, VI and VII and shall be designed to withstand not less than the hydrostatic pressures specified in table VIII. This test may be made with the hose attached in connection with test of the hose.

Table III - Dimensions for class a couplings.[1]

Name	Size	A (min.)	B	C +1/64 -0	D (min.)	E (min.)	F (min.)	G	H
Single jacket	1-1/2	3/16	[2]	1-33/64	1-1/2	3/16	1/8	1-7/8	19/32
Single jacket	2-1/2	3/16	[2]	2-33/64	1-3/4	1/4	1/4	3	15/16
Double jacket	1-1/2	3/16	[2]	1-33/64	1-1/2	3/16	1/8	1-7/8	19/32
Double jacket	2-1/2	3/16	[2]	2-33/64	1-3/4	1/4	1/4	3	15/16

[1]Unless otherwise specified in the contract or order all dimensions are in inches and plus or minus 0.010.
[2]Made to suit outside diameter of hose.
[3]On 1-1/2 inch hose dimension shall be hose o d. plus 3/32.
[4]On 2-1/2 inch hose dimension shall be hose o d. plus 5/32.

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Table III - Dimensions for class a couplings (cont'd.).[1]

K +1/32	L	M (min.)	O	P (min.)	S (min.)	T	X	Y	Z
2-1/16	5/8	1-1/4	1-1/2	0.04	1/8	3/4	1-9/16	2-1/16	1/8
3-3/16	1	1-1/2	2-1/2 or 2-7/16	.05	1/4	3/4	2-9/16	3-3/16	3/16
2-1/16	5/8	1-1/4	1-1/2	.04	1/8	3/4	1-9/16	2-1/16	1/8
3-3/16	1	1-1/2	2-1/2 or 2-7/16	.05	1/4	3/4	2-9/16	3-3/16	3/16

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Table IV - Dimensions for couplings classes b and c.[1, 2]

Class	Size	A (min.)	B	C (min.)	D (min.)	E (min.)	F (min.)	G	H
b	3/4	3/16	[3]	49/64	1-11/16	3/16	5/32	1-5/16	17/32
c	1-1/2	3/16	[3]	1-33/64	1- 3/4	1/4	1/8	1-7/8	19/32
c	2	3/16	[3]	2- 1/64	1- 3/4	1/4	1/8	2-1/2	23/32
c	2-1/2	3/16	[3]	2-33/64	1- 3/4	1/4	1/4	3	15/16
c	3	1/4	[3]	3- 1/64	2- 1/4	1/4	1/4	3-5/8	1-1 /16
c	4	1/4	[3]	4- 1/64	2- 1/4	1/4	1/4	4-3/4	1-1 /16

[1]On 3 inch and 4 inch size couplings, alternate construction using wire, pellets or ball bearing attachment of female swivel and hose bowl may be used in place of the still thread or groove shown on drawings.
[2]Unless otherwise specified all dimensions are in inches and plus or minus 0.010.
[3]To suit outside diameter of hose.

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Table IV - Dimensions for couplings classes b and c (cont'd.).[1, 2]

K +1/32	L	M	O	P (min.)	X	Y	Z
1- 7/16	9/16	1-1/4	3/4	0.040	15/16	1- 7/16	3/16
2- 1/16	5/8	1-3/8	1-1/2	.040	1- 9/16	2- 1/16	1/8
2-11/16	3/4	1-3/8	2	.040	2- 1/16	2-11/16	1/8
3- 3/16	1	1-1/2	2-1/2 or 2-7/16	.050	2- 9/16	3- 3/16	3/16
3- 3/4	1-1/8	1-7/8	3	.060	3- 1/16	3- 3/4	3/16
5- 1/8	1-1/8	2	4	.070	4- 1/16	5- 1/8	3/16

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Table V - Dimensions of threaded parts.

Nominal size	I	J	H	L	T	Threads per inch	Full threads in T
Inches	Inch	Inch	Inches	Inches	Inch		
[1]1	[2]	----	17/32	9/16	3/8	11-1/2	4-1/4
1-1/2	[2]	----	19/32	5/8	15/32	11-1/2	5-1/4
2	[2]	----	23/32	3/4	19/32	11-1/2	6-3/4
2-1/2	1/4	3/16	15/16	1	11/16	7-1/2	4-3/4
3	5/16	1/4	1- 1/16	1-1/8	13/16	6	4-1/4
4	5/16	1/4	1- 1/16	1-1/8	13/16	6	4-1/4

[1]Use for 3/4 inch, class b hose coupling.
[2]Chamfer 35 degrees.

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Table VI - Dimensions of coupling threads.

Nominal size	Threads per inch	Pitch	Depth of thread	Pitch diameter			Minor diameter - C		
				Maximum	Tolerance	Minimum	Maximum	Tolerance	Minimum
Inches		Inch	Inch	Inches	Inch	Inches	Inches	Inch	Inches
[1]1	11-1/2	0.08696	0.05648	1.2571	0.0085	1.2486	1.2091	0.0170	1.1921
1-1/2	11-1/2	.08696	.05648	1.8408	.0085	1.8323	1.7928	.0170	1.7758
2	11-1/2	.08696	.05648	2.3148	.0085	2.3063	2.2668	.0170	2.2498
2-1/2	7-1/2	.13333	.08660	3.0130	.0160	2.9970	2.9424	.0320	2.9104
3	6	.16667	.10825	3.5486	.0180	3.5306	3.4583	.0360	3.4223
4	6	.16667	.10825	4.8380	.0180	4.8200	4.7477	.0360	4.7117

[1]Use for 3/4 inch, class b hose coupling.

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Table VII - Dimensions of nipple threads.

Nominal size	Threads per inch	Pitch	Depth of thread	Pitch diameter, Maximum	Pitch diameter, Tolerance	Pitch diameter, Minimum	Major diameter - D, Maximum	Major diameter - D, Tolerance	Major diameter - D, Minimum
Inches		Inch	Inch	Inches	Inch	Inches	Inches	Inch	Inches
[1]1	11-1/2	0.08696	0.05648	1.2386	0.0085	1.2301	1.2951	0.0170	1.2781
1-1/2	11-1/2	.08696	.05648	1.8223	.0085	1.8138	1.8788	.0170	1.8618
2	11-1/2	.08696	.05648	2.2963	.0085	2.2878	2.3528	.0170	2.3358
2-1/2	7-1/2	.13333	.08660	2.9820	.0160	2.9660	3.0686	.0320	3.0366
3	6	.16667	.10825	3.5156	.0180	3.4976	3.6239	.0360	3.5879
4	6	.16667	.10825	4.7999	.0180	4.7819	4.9082	.0360	4.8722

[1]Use for 3/4 inch, class b hose coupling.

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Table VIII - Hydrostatic pressure.

Class	Hydrostatic pressure (min.)
	Pounds per square inch
a	600
b	250
c	100

3.2.2 A complete set of class a couplings shall be as shown on figure 1 and table III and shall consist of the following parts:

Part 1. Bowl of female section.
Part 2. Swivel, attached to female section by expanding the latter; swivel shall turn freely with the hand.
Part 3. Male section.
Part 4. Expansion rings, one each for male and female sections.
Part 5. Rubber washer, fitting recess in swivel.
Part 6. Rubber washer fitting recess in hose bowl of couplings for use with cotton rubber lined fire hose only. Washers in hose bowl recess to be furnished by hose in manufacturer.

3.2.3 A complete set of classes b and c couplings shall be as shown on figure 3 and table IV and shall consist of the following parts:

Part 1. Bowl of female section.
Part 2. Swivel, attached to female section by expanding the latter; swivel shall turn freely with the hand. (See [1] of table IV.)
Part 3. Male section.
Part 4. Expansion rings, one each for male and female sections.
Part 5. Rubber washer, fitting recess in swivel.

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3.2.4 The thread dimensions for all couplings as shown on figure 2 shall be as specified in tables V, VI and VII.

3.3 Finish. - A smooth sand cast finish will be acceptable on nonmating surfaces providing the fittings meet the dimensional requirements as listed herein. All mating and gasket surfaces shall be 125 r.m.s. finish or better. The inside of the outlet end of the wash deck nozzle (see figure 6) shall have a 125 r.m.s. finish for a distance of approximately 3/4 inch.

3.4. The parts for one complete nozzle type II, strainer type III and reducing coupling type IV shall be as shown on figures 4, 5 and 6 respectively.

3.5 Dimensions for the nozzle, strainer and reducing coupling shall be in accordance with figure 5 (see table IX) and figure 6 respectively. All threads shall be in accordance with tables V, VI and VII.

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3.6 Markings.- The name or trade mark of the coupling manufacturer and the size of hose for which the coupling is manufactured shall be stamped or cast on the outside of each fitting in letters at least 1/8 inch high.

Table IX - Dimensions of strainers.

Size of hose	A	B	C	D	E	F (min.)	G	H	I	J	K	L
Inches	Inches	Inches	Inches	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inch	Inches
1-1/2	1-1/2	1-15/16	2-1/4	1/8	1	1/8	11/32	3/16	1/2	1/8	15/32	6-5/16
2	2	2- 7/16	2-3/4	1/8	1	1/8	13/32	1/4	1/2	1/8	19/32	7-1/4
2-1/2	2-1/2	3- 1/8	3-3/8	1/8	1	1/4	15/32	5/16	1/2	1/8	11/16	7-7/8
3	3	3-11/16	4	1/8	1	1/4	17/32	3/8	1/2	1/8	13/16	8-5/8

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling for lot acceptance. -

4.1.1 Inspection lot. - All fittings of the same type, class and size offered for delivery at one time shall be considered a lot for purposes of inspection and test.

4.1.2 Sampling for visual and dimensional examination. - A random sample of fittings shall be selected from each lot offered for Government inspection in accordance with Standard MIL-STD-105 at inspection level II. The Acceptable Quality Level shall be equal to 1.5 percent defective.

4.1.3 Sampling for chemical analysis. - Two samples each of at least 2 ounces of drillings, millings or clippings shall be taken from separate fittings selected from each lot offered for Government test and sent to a Government laboratory for chemical analysis.

4.1.4 Sampling for hydrostatic pressure test. - A random sample of fittings shall be selected from each lot offered for Government tests in accordance with the appendix to Standard MIL-STD-105 at inspection level L-4.

4.2 Inspection and tests. -

4.2.1 Visual and dimensional examination. - Each of the sample fittings selected in accordance with 4.1.2 shall be visually and dimensionally inspected to verify compliance with the requirements of this specification.. Any fitting in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective fittings in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.2.2 Chemical analysis. - Each of the samples specified in 4.1.3 shall be separately analyzed to determine conformance with the requirements of this specification. If either sample fails to conform to the requirement, the lot represented shall be rejected.

4.2.3 Hydrostatic pressure test. - Each of the sample hose fittings selected in accordance with 4.1.4 shall be subjected to the hydrostatic pressure specified in table VIII, and the test shall be conducted or witnessed by the Government inspector, and the results of each test compared with this specification. Failure to conform to this specification shall be counted as a

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defect and the hose fitting shall be rejected. If the number of such nonconforming fittings in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection tests provided all hose fittings in the lot have been retested by the contractor and all noncompliance corrected.

4.3 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Fittings shall be packaged, packed and marked in accordance with Specification MIL-P-775, except that contact preservatives shall not be required on copper, brass, or bronze fittings.

6. NOTES

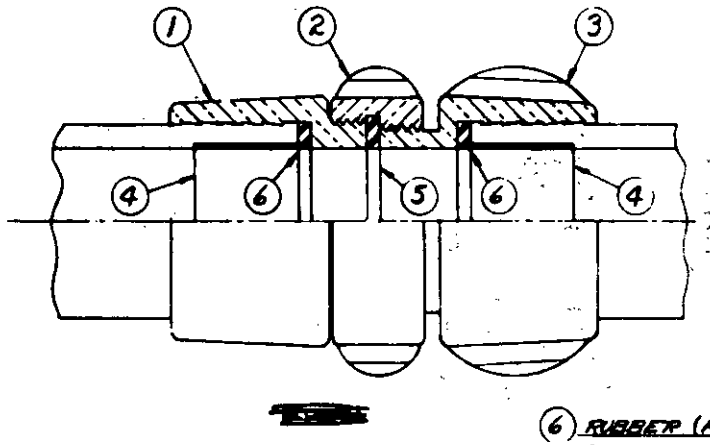
6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Size required.

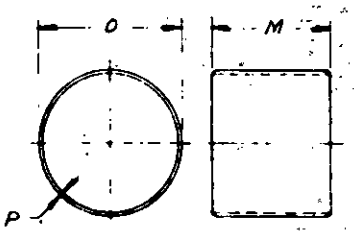
Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Bureau of Ships

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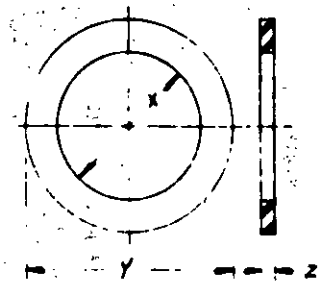


EXPANSION RING
PART 4



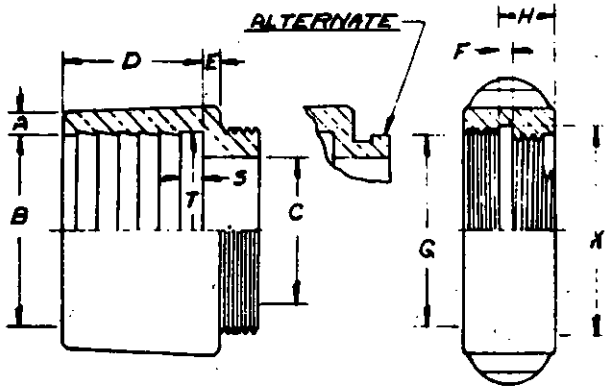
EDGES WELL ROUNDED
AND FREE FROM BURRS

RUBBER (SWIVEL) WASHER
PART 5

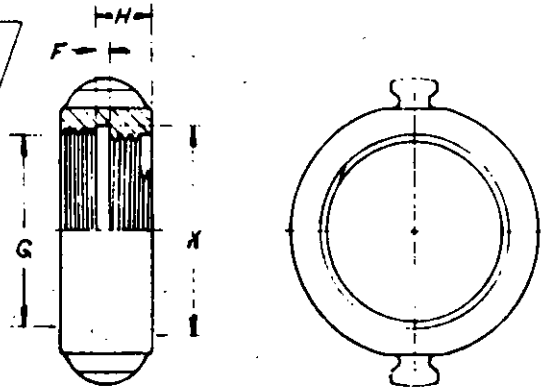


(6) RUBBER (NOSE BOWL) WASHER
SEE TO SUIT NOSE O.D.

BOWL OF FEMALE SECTION
PART 1



SWIVEL OF FEMALE SECTION
PART 2



MALE SECTION
PART 3

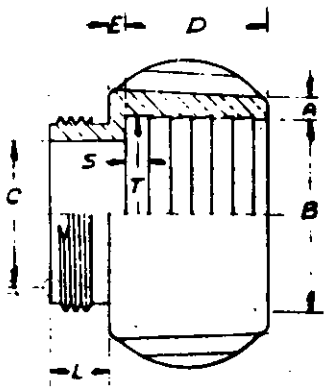
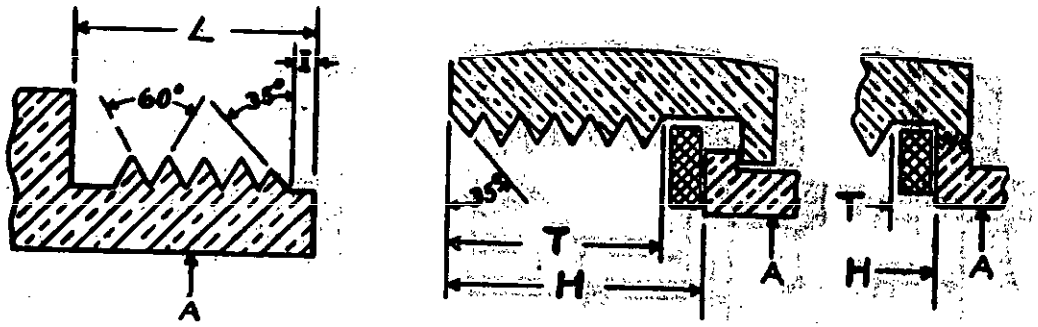


Figure 1

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Length of hose coupling screw threads for hose 2 inch diameter and below.



FOR HOSE 2½ INCHES DIAMETER AND ABOVE

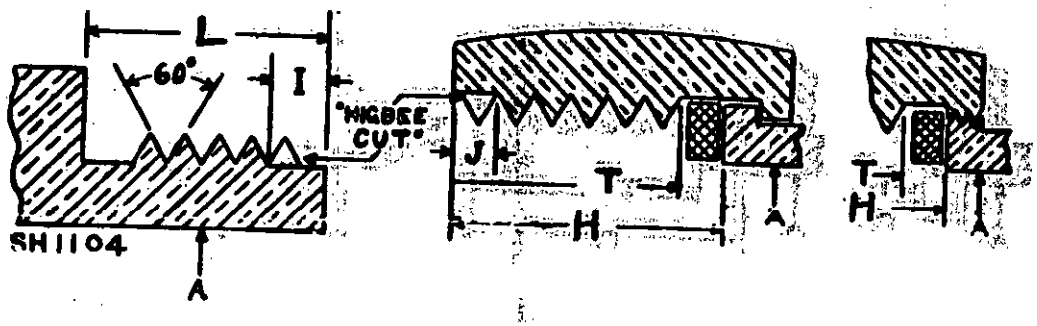


Figure 2 - Coupling threads.

(See tables V, VI, and VII for related data.)

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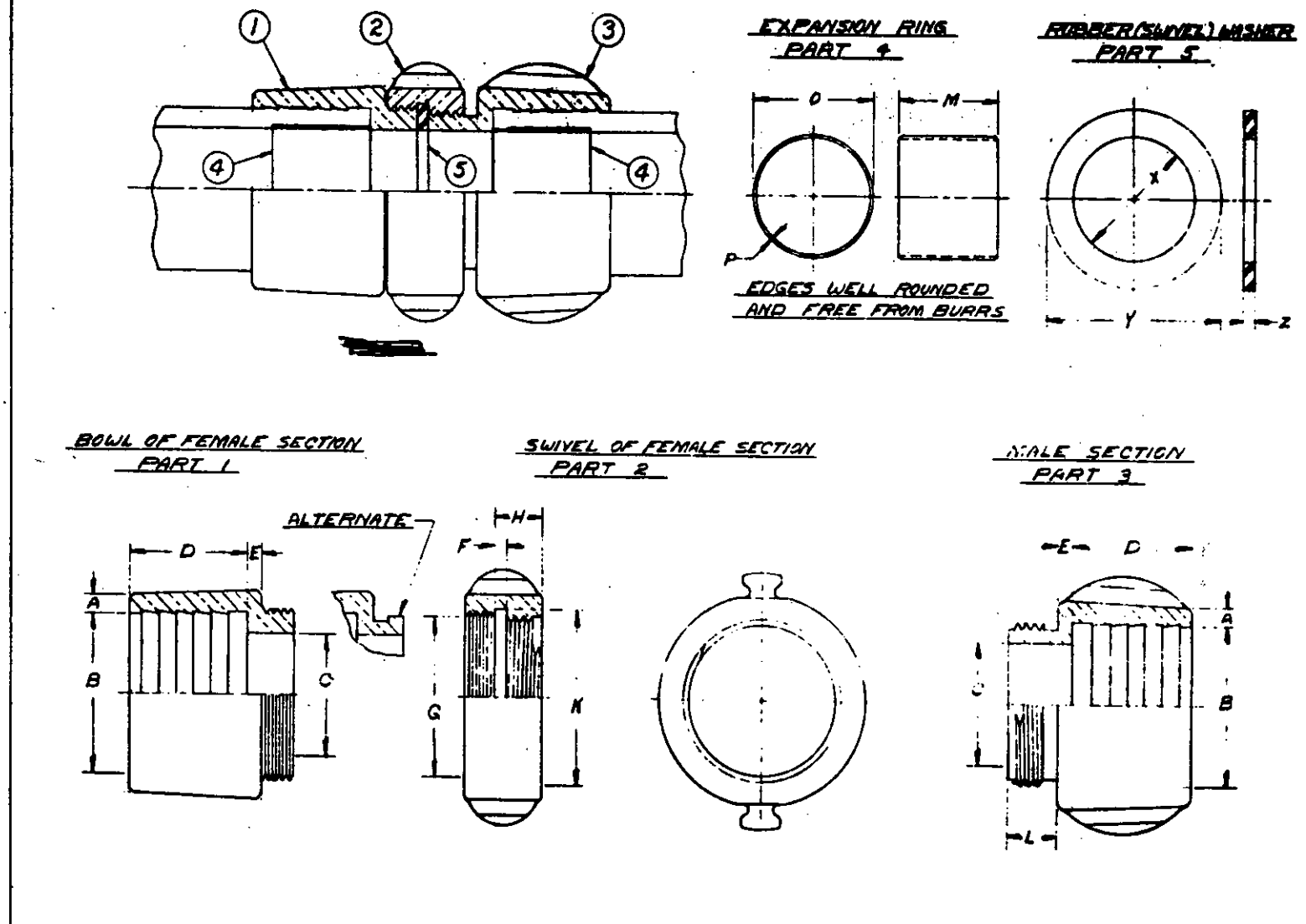
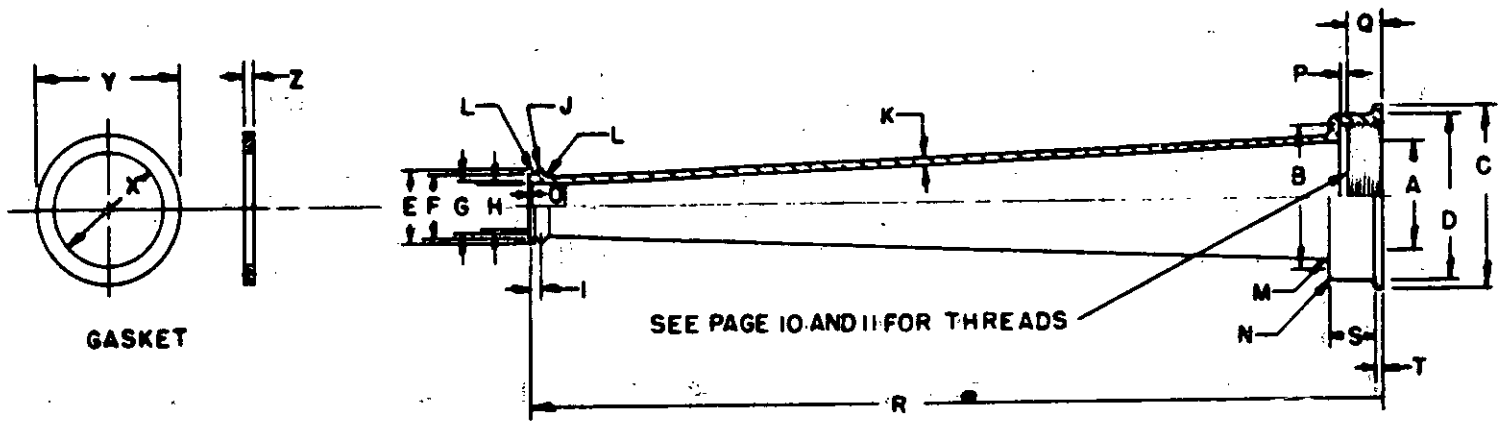


Figure 3

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DIMENSIONS OF NOZZLES

SIZE OF HOSE	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
INCHES	INCHES	INCHES	INCHES	INCHES	INCHES	INCHES	INCHES	INCH	INCH	INCH	INCH	INCH	INCH	INCH	INCH	INCH	INCH	INCHES	INCH	INCH
1-1/2	1-1/2	2-1/16	2-1/2	2-1/4	1	7/8	11/16	5/8	5/32	1/16	3/32	1/4	1/4	1/8	1/32	1/8	5/32	12	5/8	1/8
2-1/2	2-1/2	3-3/16	3-3/4	3-1/2	1-7/8	1-7/16	1-1/8	1	5/16	1/8	1/8	1/2	1/4	1/8	1/16	1/4	11/16	15	3/4	3/8

DIMENSIONS OF GASKETS

SIZE OF HOSE	X	Y	Z
INCHES	INCHES	INCHES	INCH
1-1/2	1-9/16	2-1/16	1/8
2-1/2	2-9/16	3-3/16	3/16

Figure 4 - Nozzle.

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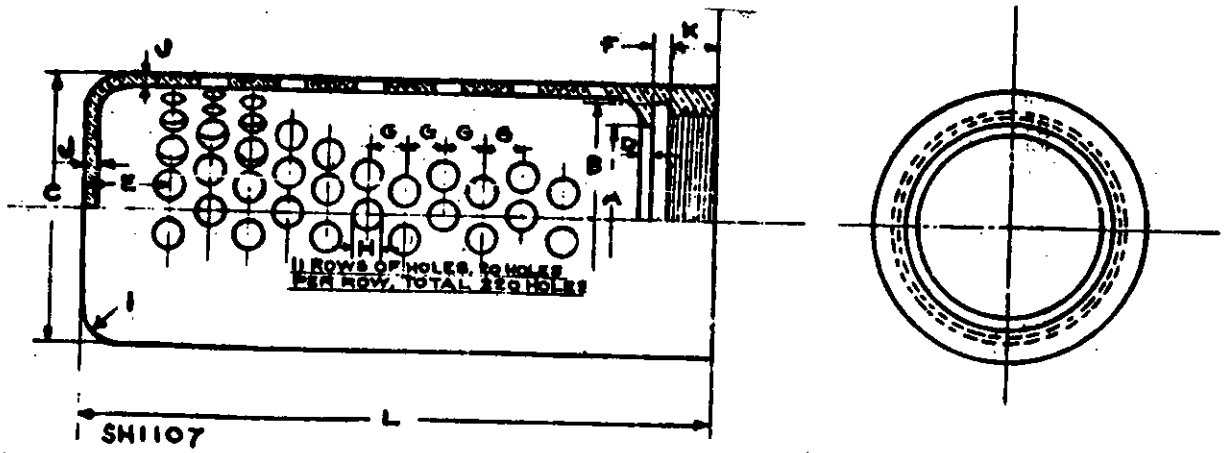
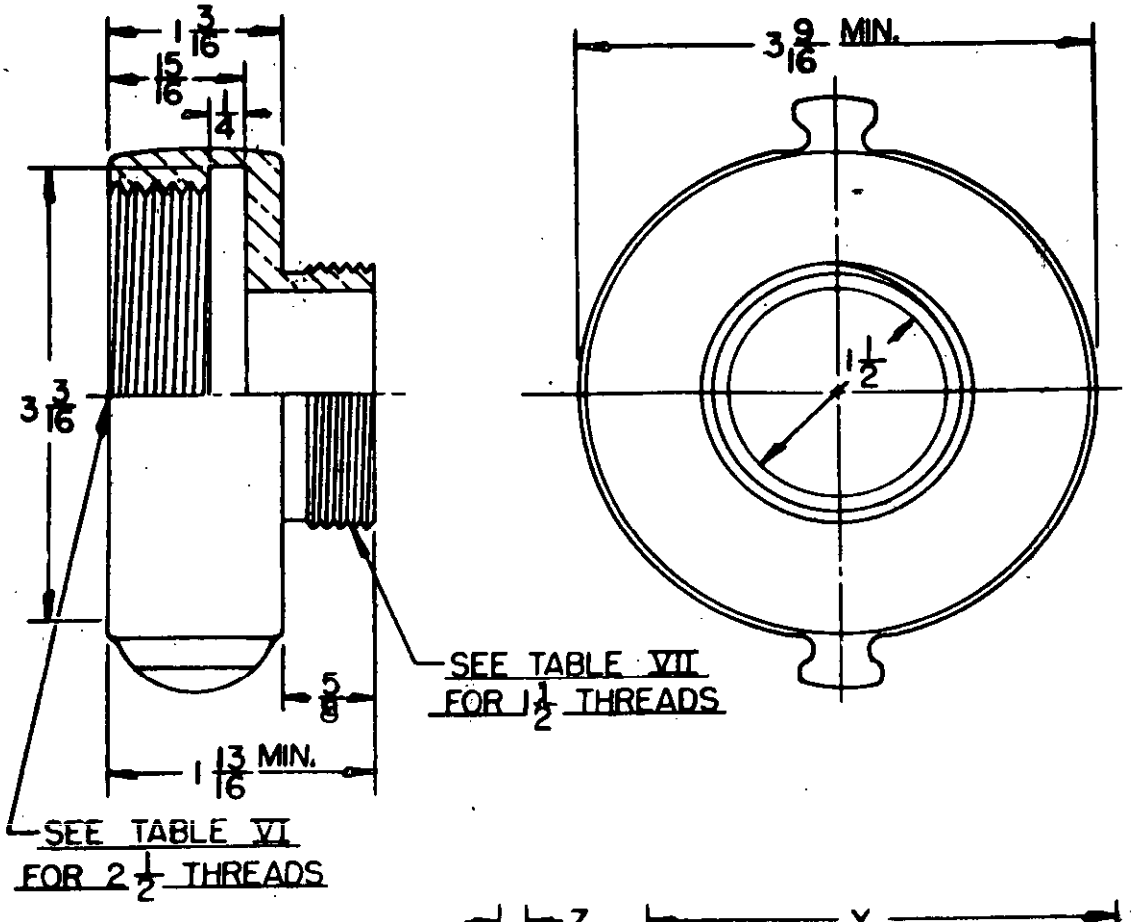
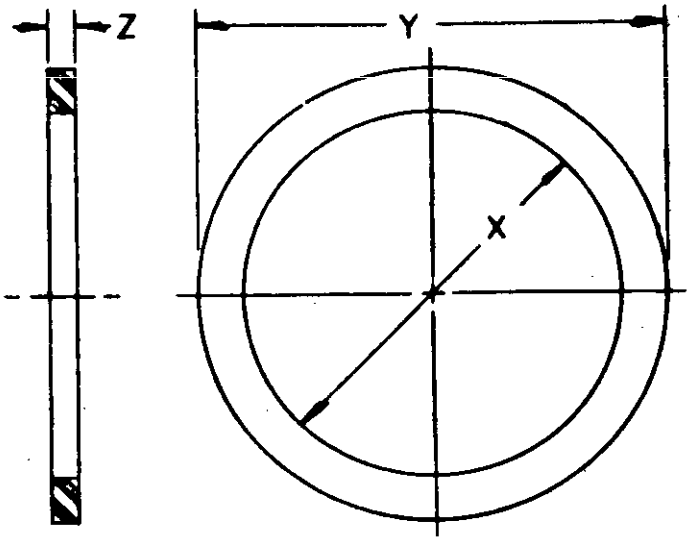


Figure 5 - Strainer.

(See table IX for related data.)

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X	Y	Z
2 9/16	3 3/16	3/16

Figure 6 - Reducing coupling.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1426
1 APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

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